Papillon Virton

Bachelor's in Engineering Sciences

papillonvirton.pro@gmail.com
Corsica · 22 y/o
github.com/PowerPeps

EDUCATION

2022 — now
Bachelor's in Engineering Sciences
in progress
Università di Corsica — Corsica

2019 — 2022
French Baccalauréat (General)
Lycée Gabriel Fauré — Ariège (France)

TECHNOLOGIES

CONCEPTS

PARADIGMS

OOP · Functional · Event-driven

ALGORITHMS

Data Structures · Complexity · Compilation

SYSTEMS

Embedded · Hardware Arch. · Low-level

DATA

SQL / Databases · Geostatistics · GIS / Mapping

FABRICATION

DAW · CAD / CAM

INTERESTS

Chess · Motorbike · Tabletop RPGs · Snowboard · Blacksmithing · Woodworking / Carpentry · Electronics · Computer Science · Open-Source

ABOUT

Studying for a Bachelor's in Engineering Sciences, with an interest in computers and electronics from an early age. Self-taught experience in exploring and understanding technical systems, through experimentation and disassembling devices to analyze how they work. Driven by personal experimentation ranging from hardware projects controlled by microcontrollers or single-board computers (Arduino, Raspberry Pi) to learning and using several programming languages (PHP, JavaScript, MySQL, Python, C#, C++), with a particular interest in the interaction between hardware and software and the design of technical systems. Having served as class representative, student representative on the school board, boarding student representative, and member of the student council provided opportunities to develop skills in representing a group, structuring information, and coordinating exchanges between students and institutions.

EXPERIENCE

2024 — **Developer/Analyst: Air Quality Data — QualitairCorse** — Contract

• Development of a web tool for visualizing and exporting air quality measurement results, featuring an interactive map interface with dynamic display, multi-criteria filtering, and geospatial data export. • Management of radon measurement campaign results: processing and validation of sensor data, statistical and temporal analyses, concentration modeling, and production of maps and exposure indicators. • Design and calibration of geostatistical models (experimental variogram analysis, theoretical model fitting, ordinary kriging, cross-validation) for predictive spatial modeling of radon concentrations.

Laravel · Livewire · GIS / Carto

2019 — **Introduction to Digital Fabrication Machines — Fablab Arize Leze** — internship

Internship at a business hotel and Fablab. Training and hands-on use of digital fabrication tools: CNC router, FDM 3D printer, vinyl cutter, and laser cutter/engraver. Learned associated CAD software and manufacturing workflows.

CAD

2019 — 2022 — **Computer Equipment Repair and Refurbishment — Ressourcerie Zero Neuf** — volunteer

Regular volunteering in Les Bordes-sur-Arize. Repair and refurbishment of household appliances, computer equipment, and more.

Repair

2017 — **Introduction to Wood Machining and Craft — Atelier Renard** — internship

Introduction to wood turning. Chuck and between-centres techniques.

Summer 2015 — **Electrical Compliance Work — La Bouguetoise** — volunteer

Rehabilitation of a community dwelling in Bouguet (09290). Electrical compliance work and alternative masonry (earth-straw eco-construction techniques).

2012 — 2021 — **Programming and Electronics Training — Private Tutoring** — training

• Design and prototyping of electronic circuits. • Embedded software development. • Component soldering and prototype assembly. • Low-level programming on microcontrollers (C++/Arduino). • Hardware-software integration and electronic system debugging.

Electronics · Computer Science

2011 — 2017 — **Lead Sound and Lighting Technician — Festival Terre-de-Couleurs** — volunteer

Sound and lighting engineering for the band "Les Tzigales" and various musical projects and jam sessions, including the Terre-De-Couleurs festival.

PROJECTS

Alpenstock public — Collection of generic directives for Alpine.js

A library of Alpine.js directives focused on ease of use.

Oforge in dev — Visual Novel Engine

A fully-featured Visual Novel engine. Scripting system, scene rendering, dialogue management, and narrative branching.

Varablizer in dev — Stack-based compiler and executor (LIFO)

Stack-based compiler and executor. Lexical analysis, bytecode compilation, and execution on a stack-based virtual machine. Developed for a Compilers course (3rd-year Bachelor's).

Alea Jacta Est in dev — Card game — Tarot de Marseille × Balatro

Card game based on the Tarot de Marseille, built with MonoGame as a university group project.

Thallium AI / NLP — Discord bot powered by Transformers and LLMs

Intelligent Discord bot using Transformer and LLM architectures. Natural language understanding, contextual response generation, and conversational interactions.